The Anatomy of a Successful Restaurant: How Business Features and Engagement on Yelp Drive High Ratings

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***Abstract*—Understanding what makes a restaurant successful on Yelp can empower businesses to improve customer satisfaction and drive engagement. This study leverages the Yelp Open Dataset to identify business features and customer engagement metrics that correlate with high ratings and review counts. By analyzing trends in restaurant attributes, such as takeout services, reservations, and ambiance, this research provides actionable insights for businesses seeking to enhance their presence on Yelp. This study employs SQL, Python, and Tableau for data extraction, cleaning, and visualization, offering a foundation for further exploration into restaurant success.**

***Index Terms*—data analysis, yelp dataset, online reviews, restaurant businesses**

I. Introduction

A. About Yelp

Yelp is a widely-used online platform that connects consumers with local businesses through crowd-sourced reviews and ratings. For restaurants, Yelp serves as both a discovery tool for customers and a critical resource for businesses aiming to improve their visibility and reputation. The platform allows users to rate businesses on a five-star scale and provide detailed reviews, which collectively shape public perception and influence consumer choices. In addition to reviews, Yelp offers business owners tools to engage with customers, such as responding to feedback, updating business information, and accessing analytics. This combined focus on user feedback and business improvement makes Yelp an ideal dataset for analyzing trends in customer engagement and identifying attributes that drive high ratings and review counts, offering valuable insights for restaurant success.

B. Problem

While Yelp provides tools like the Yelp Blog for Businesses to help improve customer satisfaction, the exact factors distinguishing highly-rated restaurants remain unclear. High star ratings signal quality and popularity, influencing customer decisions, while customer engagement, measured by review volume, serves as another key metric. However, the diversity of restaurants and customer preferences complicates identifying universal success factors.

Identifying actionable strategies based on measurable attributes is essential for restaurants in a competitive landscape. Features like delivery, takeout, ambiance, and customer service, alongside engagement behaviors such as responding to feedback and maintaining accurate information, can shape customer perceptions. This study utilizes the Yelp Open Dataset to explore correlations between restaurant attributes, engagement metrics, and their impacts on ratings and reviews, providing data-driven insights for restaurant owners to enhance success.

C. Objective

This project analyzes the Yelp Open Dataset to uncover patterns and trends that guide restaurants in improving customer engagement and ratings. High ratings and review counts on Yelp enhance visibility and customer trust, driving business growth. By identifying key attributes and engagement behaviors linked to success, this study provides actionable insights to boost customer satisfaction and loyalty.

Restaurants were selected due to their prominence as the second-largest category on Yelp and their universal relevance. They offer a practical lens for exploring customer preferences and business performance, with rela-

tively standardized factors like cuisine type and service quality, ensuring findings are broadly applicable across small diners and large chains.

Data analysis leveraged SQL for querying, Python (Pandas) for preprocessing, and Tableau for creating visualizations. This combination facilitated the extraction and visualization of meaningful correlations between business attributes, engagement metrics, and success indicators, offering strategies for restaurant owners to excel in a competitive market.

II. Related Work

The Yelp dataset has been the focus of various studies exploring its potential for understanding business success, user engagement, and review dynamics. This section summarizes notable works that provide context and inspiration for this study.

Hegde et al. [1] analyzed high-frequency business attributes and crowd dynamics to assist entrepreneurs in setting up new restaurant businesses. Their work highlights critical features such as credit card acceptance and desired ambiances like "divey" that resonate with customers. The study also employs the k-d tree algorithm to locate nearby competing restaurants, providing actionable insights for business placement and strategy.

Cervellini et al. [2] examined "trendsetters" within the Yelp dataset: users who identify and promote emerging businesses early. By analyzing tip frequencies and social graph metrics such as eigenvector centrality and PageRank, the study identified key influencers. This user-centric perspective contrasts with business-focused analyses, emphasizing the role of individuals in driving popularity through early adoption and recommendation.

Rafay et al. [3] developed a robust sentiment analysis framework for predicting review ratings using machine learning and deep learning techniques, including Naive Bayes and convolutional Long Short-Term Memory (CLSTM) models. Their findings demonstrate the importance of sentiment-based analysis in understanding customer feedback and highlight biases in traditional rating mechanisms.

Asghar [4] treated review rating prediction as a multi-class classification problem, combining various feature extraction techniques (e.g., unigrams, bigrams, trigrams) with supervised learning algorithms such as logistic regression and Naive Bayes. Their work emphasizes the importance of textual features in predicting star ratings, providing a framework for analyzing review content.

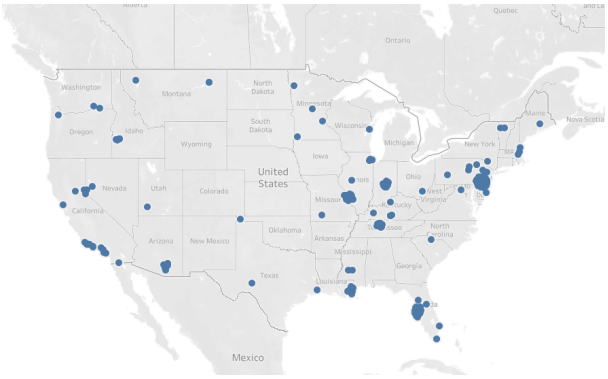

Fig. 1. Map of United States cities appearing in the Yelp Open Dataset.

Cui [5] evaluated the structural dynamics of the Yelp social network, including user clustering and the six degrees of separation theory. Insights into the network's connected components and user interactions offer valuable perspectives on information propagation and influence within Yelp.

These studies collectively underscore the multifaceted nature of the Yelp dataset, ranging from business attribute analysis and user influence to advanced machine learning applications. This research builds on these foundations by focusing on restaurant success factors, incorporating both business features and customer engagement metrics to provide actionable recommendations for improvement.

III. Dataset

The Yelp Open Dataset is a rich collection of information designed for academic and educational use, providing an extensive resource for analyzing business dynamics and customer engagement. It consists of five JSON files: `business`, `review`, `user`, `tip`, and `checkin`. Each file serves a specific purpose, offering unique insights into different aspects of the Yelp ecosystem. The dataset, while comprehensive, is an extract limited to a subset of cities primarily in North America (Figure 1). This limitation provides a snapshot of Yelp's data but does not encompass its entire global user base.

The `business` file is the cornerstone of this study, containing details about each business, such as its unique ID, name, location, star rating, and review count. Additionally, it provides a rich set of attributes that describe various features of the business, such as whether it offers takeout (`RestaurantsTakeOut`), has outdoor seating (`OutdoorSeating`), or is wheelchair accessible (`WheelchairAccessible`). For example, the

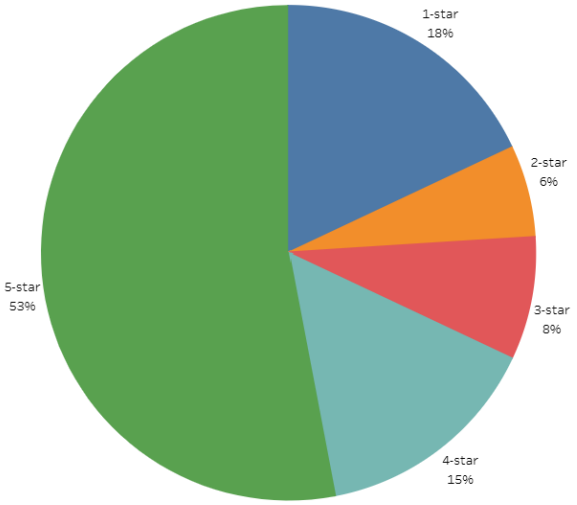

Fig. 2. Distribution of Yelp star ratings for reviews.

`Ambiance` attribute allows multiple descriptors like "romantic" or "trendy," each marked as `true` or `false`. This file also includes the `categories` field, which lists all the business categories applicable to each business, such as `Restaurant` or `Mexican`.

The `review` file complements the `business` file by providing detailed information about individual customer reviews. Each review is linked to a specific business and user via unique IDs and includes the star rating, the textual content of the review, and counts for reactions such as `useful`, `funny`, and `cool`. The dataset also features additional files like `user`, which contains profile information about Yelp users, such as their total number of reviews and average star rating; `tip`, which includes brief customer recommendations or observations about businesses; and `checkin`, which tracks the time and date of customer visits. Together, these files offer a multi-dimensional view of Yelp's data, enabling nuanced analysis of business performance, customer behavior, and the factors that drive success on the platform.

IV. Analysis

A. Star Ratings Distribution

Initial analysis, seen in Figure 2, revealed that most reviews on Yelp tend to be positive, with the majority rated 5 stars. This suggests customers are more likely to leave reviews after positive experiences. On the other end, the second most common star rating is 1 star, which indicates customer are also likely to leave reviews after negative experiences.

B. Business Ratings Distribution

The pie chart in Figure 3 illustrates the distribution of star ratings provided by Yelp users for businesses. The data reveals the following key insights:

- **Dominance of Positive Ratings:** The majority of ratings are 4-star (38%) and 3-star (24%), accounting for over 60% of businesses. This indicates that users are generally inclined to give favorable or average ratings, reflecting a bias toward positive experiences.
- **Relatively Low Negative Ratings:** Negative reviews (1-star and 2-star) make up a smaller proportion of the overall ratings, with 1-star reviews accounting for only 5% and 2-star reviews 16%. This suggests that customers are less likely to leave extremely negative feedback unless they had a particularly bad experience.
- **Distribution Highlights User Behavior:** Users appear more likely to leave a review after a positive or neutral experience rather than a highly negative one. This aligns with common user behavior patterns on review platforms, where satisfied customers tend to share their opinions more frequently.
- **Implications for Businesses:** Businesses aiming to improve their Yelp ratings should focus on strategies that encourage customers to leave positive feedback. By ensuring quality service and addressing key areas of concern (e.g., customer service, cleanliness, and accurate business information), businesses can shift their ratings distribution toward higher stars. Additionally, targeting customers who have had positive experiences with incentives for reviews could further increase the proportion of 4-star and 5-star ratings.

This analysis underscores the importance of understanding customer sentiment and encouraging feedback as a tool for improving ratings and business performance.

C. Categories

The raw dataset contains 1,311 unique business categories. Analyzing such a large and varied set of categories required processing and clustering to identify patterns and group similar categories into broader, meaningful labels. By employing Python and k-means clustering, the dataset was distilled into actionable insights that capture the essence of category representation and engagement on Yelp.

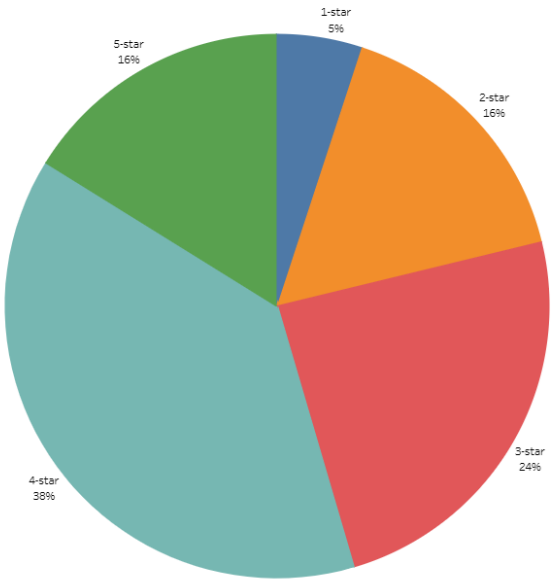

Fig. 3. Distribution of Yelp star ratings for businesses.

- **Exploding Categories:** Since each business could be associated with multiple categories (stored as a comma-separated string in the `categories` column), it was necessary to split these strings and treat each category as an individual data point. Using Pandas, the `str.split` and `explode` methods were employed to separate and flatten the categories, resulting in a larger dataset where each row corresponds to a single category instance.
- **Counting Category Frequencies:** After exploding the categories, the occurrences of each unique category were counted using the `value_counts` method. This provided the raw frequency of each category in the dataset, which was then exported to a CSV file for further analysis or visualization.
- **Visualization:** Figure 4 was created using Tableau, which provided an intuitive way to display the top business categories (e.g., `Restaurants`, `Shopping`) by their frequency. This visualization highlights the dominance of restaurants as the largest category on Yelp, followed by other common categories such as `Shopping` and `Home & Local Services`.

Clustering Categories

- **Clustering with k-Means:** The k-means clustering algorithm was applied to group the categories into 10 clusters, representing broader groups of business types. Each cluster was then manually mapped to a more meaningful label, such as `Restaurants`, `Shopping`, and `Beauty & Fitness`.

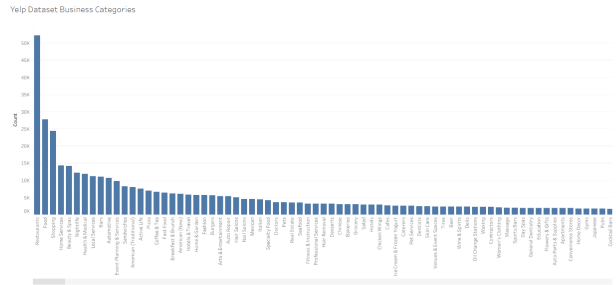

Fig. 4. Extracted distribution of top business categories on Yelp.

- **Mapped Labels and Aggregation:** The results of the k-means clustering were added back to the dataset, allowing each original category to be associated with its broader mapped label. The frequency of each broader category was then calculated, leading to a more concise representation of the dataset.
- **Visualization:** Figure 5, showing reviewed businesses by category (e.g., `Home & Local Services` at 20% and `Restaurants` at 17%), was created using Tableau. This chart highlights the proportion of reviews attributed to each of the mapped, high-level categories, providing a clear view of customer engagement across business types.

1) Insights from the Analysis:

- **Dominance of Restaurants:** Restaurants dominate both the raw category distribution and the percentage of reviews, reflecting their central role in Yelp's ecosystem. This underscores the relevance of focusing on restaurants for understanding success metrics.
- **Relevance of Broader Categories:** Mapping over 1,300 unique categories into 10 broader groups simplifies the analysis and provides actionable insights. For example, high review engagement in categories like `Home & Local Services` and `Shopping` suggests a strong customer interest in these sectors.
- **Practical Applications:** The clustering and visualization process provides a blueprint for businesses to understand their relative positioning within broader groups and focus on features that drive higher engagement within their category.

By leveraging k-means clustering, and Tableau, this analysis transforms raw categorical data into digestible

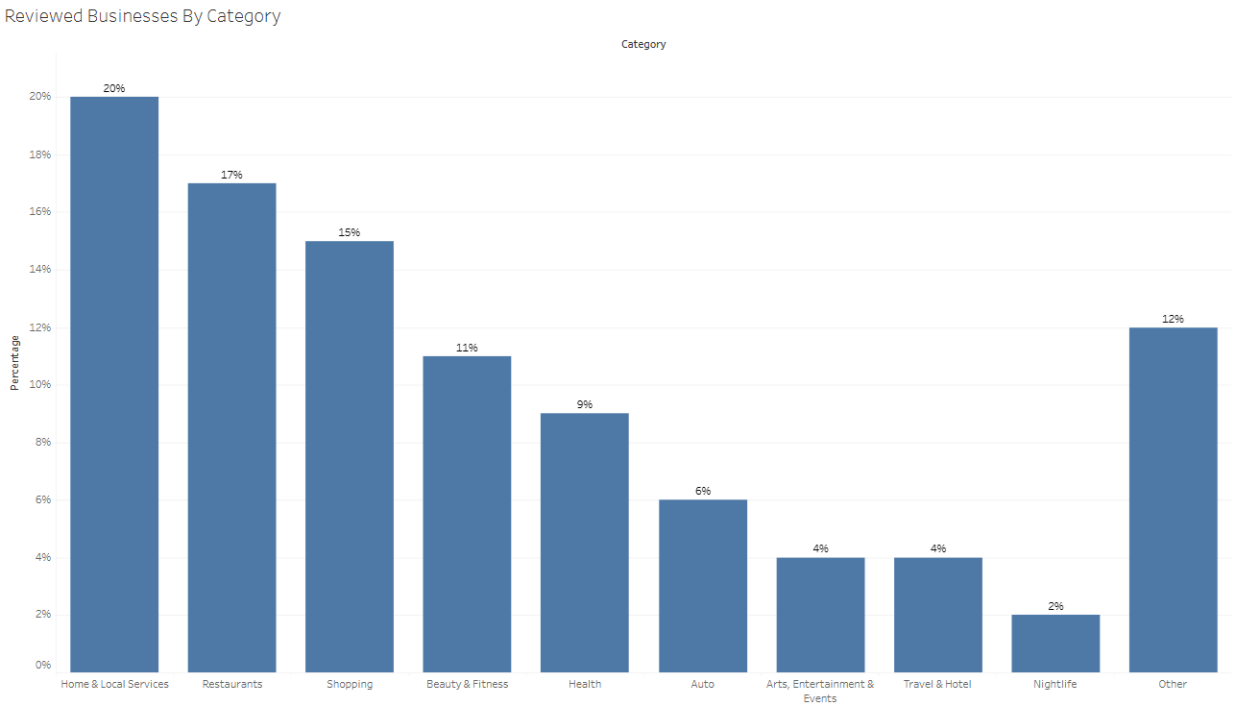

Fig. 5. Reviewed businesses by clustered categories.

insights that highlight customer engagement trends and category significance on Yelp.

D. Business Attributes

The attributes column in the dataset included a variety of features that describe business characteristics, such as whether the establishment accepts credit cards, provides takeout or delivery services, has outdoor seating, or allows pets. These attributes were represented in different formats, including **boolean** (e.g., `True` or `False`), **categorical** (e.g., types of alcohol served), **nested** structures (e.g., ambiance), and **numerical** values (e.g., price range).

1) Exploring Select Business Attributes: To further explore the dataset and its significance, this graph provides an extract of three select business attributes - WiFi availability, alcohol options, and outdoor seating - and their correlations with average review counts and star ratings. These attributes were chosen to provide insights into their potential impact on customer preferences and business success.

- **WiFi:** Businesses offering free WiFi tend to have slightly higher average review counts compared to those with paid WiFi or no WiFi at all. However, the average star ratings remain consistent across all categories, suggesting that while free WiFi may attract more engagement, it does not significantly influence customer satisfaction.
- **Alcohol:** Establishments serving alcohol (e.g., beer and wine or a full bar) receive significantly higher average review counts compared to those that do not serve alcohol. This finding highlights the importance of alcohol as a value-adding feature that encourages customer engagement. Interestingly, the average star ratings for all three categories - beer and wine, full bar, and no alcohol - remain comparable, indicating that while alcohol increases customer interaction, it does not necessarily enhance overall ratings.
- **Outdoor Seating:** Restaurants with outdoor seating show considerably higher average review counts compared to those without this feature. Additionally, the average star ratings for businesses with outdoor seating are slightly higher, indicating that outdoor seating may enhance both engagement and satisfaction. This aligns with the broader findings in the dataset, which suggest that convenience and ambiance are key drivers of customer preferences.

These visualizations illustrate how specific business attributes can influence customer interaction and satisfaction. By analyzing these features, businesses can better tailor their offerings to align with customer expec-

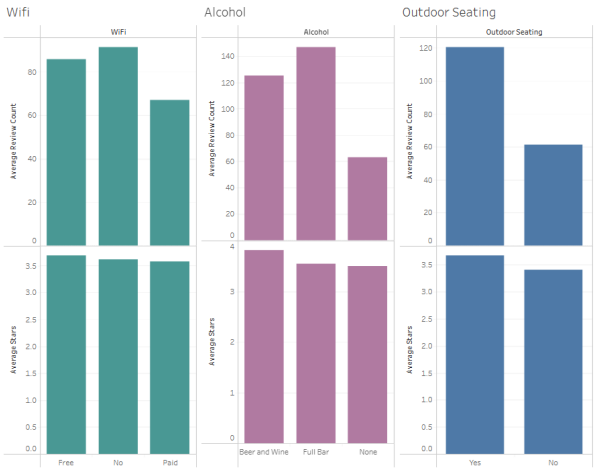

Fig. 6. Select business attributes plotted on bar graphs to faciliate understanding.

tations and potentially improve both their Yelp presence and overall success.

E. Business Attributes and Correlations

For the analysis, these attributes were normalized into individual columns to enable direct correlation analysis.

To process the data, boolean attributes were converted into binary values (1 for `True`, 0 for `False`), and categorical variables such as `Alcohol` (e.g., `none`, `beer and wine`, `full bar`) and `RestaurantsPriceRange` were mapped to numerical scales. This allowed for the calculation of correlation coefficients between business attributes and two key metrics: star ratings and review counts.

1) Correlation Analysis: The correlation graph in Figure 7 illustrates the relationship between various business attributes and their impact on star ratings and review counts. Key insights from the analysis include:

- **Positive Correlations with Review Counts:** Attributes such as `RestaurantsGoodForGroups`, `RestaurantsTableService`, and `OutdoorSeating` exhibit strong positive correlations with the number of reviews. This indicates that businesses offering these amenities are more likely to engage customers and encourage feedback.
- **Negative Correlations with Star Ratings:** Interestingly, the presence of a `DriveThru` negatively correlates with star ratings, suggesting that businesses with drive-thrus may prioritize convenience over quality, potentially leading to less favorable reviews.
- **Limited Impact of Certain Attributes:** Some attributes, such as `BikeParking` and `HasTV`, show minimal correlation with either metric, suggesting that these features may not significantly influence customer behavior.

2) Implications for Businesses: The analysis highlights the importance of prioritizing high-impact features to improve both ratings and customer engagement. Restaurants should consider focusing on attributes that enhance customer experience, such as group-friendly seating, ambiance, and outdoor options, to encourage positive reviews and higher star ratings.

V. Implications

The findings from this study provide actionable strategies that restaurants can implement to improve both their Yelp ratings and customer engagement. By leveraging the insights gained from the analysis, businesses can focus on the features and attributes that are most impactful for customer satisfaction and engagement.

A. Focus on Convenience

Restaurants that prioritize convenience tend to perform better in terms of both reviews and ratings. Features such as delivery, takeout, and outdoor seating are strongly correlated with customer engagement. These attributes cater to modern customer preferences for flexibility and accessibility. By investing in these services, restaurants can appeal to a broader audience, including those who value convenience over traditional dining experiences.

B. Enhance Ambiance and Atmosphere

Attributes related to ambiance, such as romantic settings, suitability for groups, and outdoor seating, are positively associated with higher ratings. Customers often seek dining experiences that go beyond food quality to include the environment and atmosphere. Restaurants should consider investing in decor, lighting, and seating arrangements to create a welcoming and memorable environment that encourages customers to leave positive reviews.

C. Prioritize High-Impact Features

Not all attributes are equally influential. Features such as table service, group-friendly seating, and availability of alcohol are significantly correlated with positive ratings and reviews. On the other hand, features like drive-thrus and low price ranges tend to correlate negatively with star ratings. Restaurants should prioritize

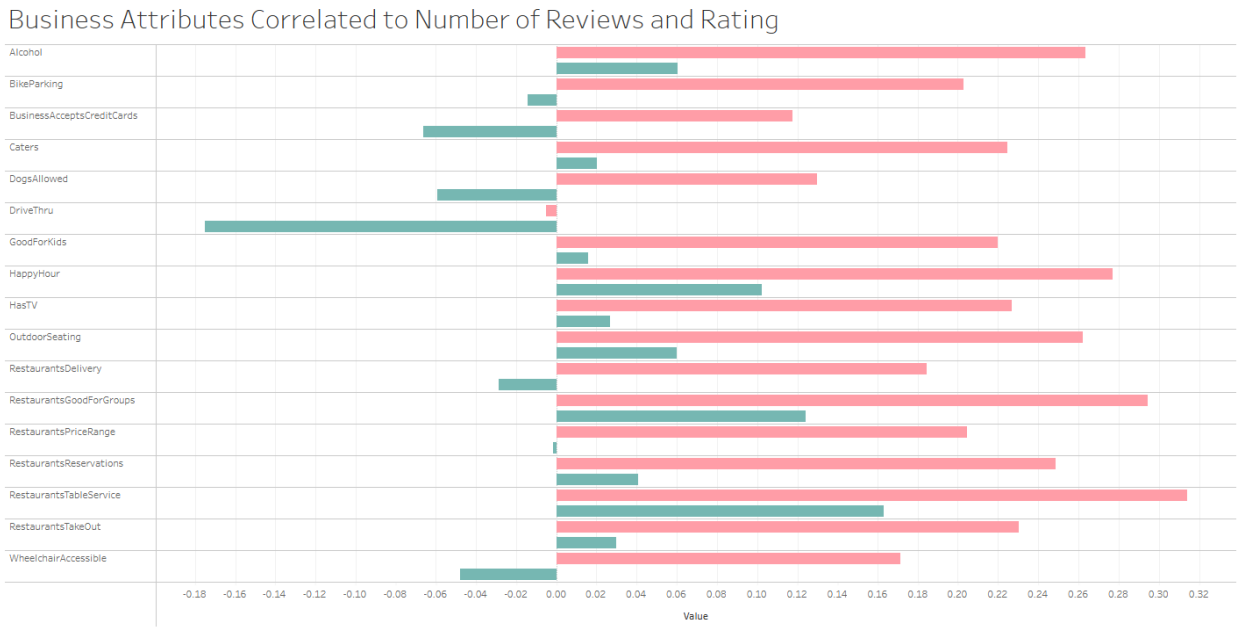

Fig. 7. Business Attributes Correlated to Number of Reviews and Ratings. Correlations are shown in pink (reviews) and teal (ratings).

implementing features that enhance the dining experience and align with their target customer base, while avoiding over-reliance on features that may detract from perceived quality.

D. Encourage Positive Reviews

Since the majority of Yelp reviews tend to be positive, restaurants can benefit from encouraging satisfied customers to leave reviews. Strategies such as loyalty rewards, discounts for repeat visits, or friendly reminders after meals can help increase review counts and improve ratings.

By focusing on these actionable strategies, restaurants can effectively enhance their presence on Yelp, improve customer satisfaction, and ultimately drive business success.

VI. Further Work

The findings of this study lay the groundwork for several future research directions that could provide deeper insights into the factors driving restaurant success. By extending the scope of analysis, future studies could uncover additional trends and actionable strategies.

A. Temporal Analysis

Future work could explore how ratings and reviews evolve over time or vary by time of day or day of week. For instance, analyzing review trends during holidays, weekends, or special events could reveal how external factors impact customer satisfaction and engagement. Temporal patterns could also help restaurants optimize their operations, such as staffing during peak hours or promoting specific offerings during slower periods.

B. Spatial Analysis

A spatial dimension of analysis could compare restaurant success across different geographic contexts, such as urban, suburban, and rural establishments. Investigating regional or international differences could also uncover how cultural preferences influence customer expectations and ratings. For example, a comparative study of Yelp's popularity in North America versus other countries might highlight differences in how customers engage with review platforms.

C. Textual Analysis

Textual analysis, such as sentiment analysis, could be used to examine the language of customer reviews. This approach could identify key themes, recurring complaints, or highly valued attributes. Sentiment trends over time could reveal how public perception of a business changes, while deeper linguistic analysis could highlight differences in review tone between high-performing and low-performing restaurants. For example, identifying frequently mentioned words like "trendy" or "family-friendly" in specific locations could help businesses tailor their marketing and offerings.

D. Ordinal Analysis

Ordinal analysis could explore the impact of rating gradients, such as "$," "$$," "$$$," or "$$$$," on customer perceptions and business success. Investigating the relationship between price level and average ratings could provide insights into how customers perceive value for money. Similarly, understanding how attributes rated as "good," "better," or "best" correlate with ratings and review counts could help restaurants prioritize service improvements.

E. Conclusion

Further work in these areas could provide a more comprehensive understanding of the factors driving restaurant success on Yelp. By leveraging temporal, spatial, textual, and ordinal analyses, researchers can uncover nuanced trends and actionable insights to guide restaurants in improving their performance and customer engagement.

VII. Conclusion

This study analyzed the Yelp Open Dataset to identify key factors contributing to the success of restaurant businesses on the platform. By examining business attributes such as takeout services, ambiance, and customer engagement metrics like review volume and response rates, we uncovered actionable insights for restaurant owners looking to improve their Yelp presence. The findings suggest that features like outdoor seating, alcohol availability, and group-friendly seating significantly influence customer engagement and ratings. Additionally, restaurants that prioritize convenience and ambiance are more likely to receive positive reviews. The analysis provides a solid foundation for businesses to refine their offerings and engage more effectively with customers, enhancing their success on Yelp and driving growth. Future work could expand on these findings by exploring temporal, spatial, and textual patterns to uncover deeper insights into customer behavior and restaurant success.

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